

Two specificity markers *etten* ‘some’ and *han* ‘one’ in Korean and their implications[‡]

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Abstract

In this paper I discuss two markers for specificity, *etten* and *han*, in Korean. First, I discuss their structural properties and claim that *etten* is a determiner, and *han* is a numeral preceded by an implicit existential quantifier. Then I discuss four uses of *etten* and two uses of *han*, and show when they are used as specificity markers. There are various properties of the two specificity markers, some of which they have in common. I take them as properties of specificity markers in general. The properties only one of them has are taken to come from the original meanings of the two morphemes. I claim that specific indefinites range over individual concepts, which lead to the properties they have in common.

1 Introduction

There have been a lot of studies on specific indefinites in various languages, but specific indefinites show different syntactic or semantic characteristics in

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[‡]acc(usative), adn(ominal), cl(a)s(sifier), c(o)mp(lementizer), dec(larative), del(imiter), hon(orific), imp(erative), int(errogative), mod(ality), n(o)m(ina)l(izer), nom(inative), n(on)p(a)st, p(a)st, pl(ural), top(ic)

different languages. One reason is that when a morpheme is used to express specificity, it is often co-opted from some morpheme which expresses something related to, but actually different from, specificity. In these cases, we have to find out ways to distinguish uses of the morpheme, whether they are used as specificity markers or as its original uses. This is what I am going to do about Korean.

In this paper I discuss two markers for specificity, *etten* and *han*, in Korean. First, I discuss their structural properties and claim that *etten* is a determiner, and *han* is a numeral preceded by an implicit existential quantifier. Then I discuss four uses of *etten* and two uses of *han*, and show when they are used as specificity markers. I take the properties of the two specificity markers they have in common as properties of specificity markers in general. The properties peculiar to either of them are likely to come from the original meanings of the two morphemes. And to account for the properties of specific indefinites, I claim that specific indefinites range over individual concepts.

To show that the two morphemes are used as specificity markers, we need to compare two contexts in which specific and nonspecific NPs show contrasts.

- (1) mino-nun yeca-lul manna-ss-ta. ku yeca-nun
Mino-TOP woman-ACC meet-PST-DEC. the woman-TOP
yeppu-ess-ta.
pretty-PST-DEC
Mino met a woman. The woman was pretty.

It is well-known that indefinites without any specificity markers can be used as specific indefinites. In this example, the primary meaning of *yeca* is an arbitrary woman. In this respect, the indefinite can be taken to be nonspecific. However, it can be referred to by a pronoun in a later discourse. I suppose that this is the effect of the use of a pronoun. The first sentence introduces a woman Mino met. The specific event of Mino's meeting a woman selects a specific woman indirectly.¹

- (2) ??mino-nun yeca-ka phiyoha-ess-ta. ku yeca-nun yeppu-ess-ta.

¹The NP can be construed as definite, and can be referred to by a pronoun. But we do not consider this reading.

Mino-TOP woman-ACC need-PST-DEC. the woman-TOP pretty-PST-DEC

Mino needed a woman. The woman was pretty.

In contrast with the verb *manna* ‘meet’, the verb *philyoha* ‘need’ takes an object which is referentially opaque if it is a nonspecific indefinite. For this reason, it is not possible for the definite NP *ku yeca* ‘the woman’ to refer to the woman in the first sentence.

This can be compared with an indefinite with *etten* or *han*. In (3) the NP with *etten/han* can be referred to by *ku yeca*.

- (3) Mino-nun etten/han yeca-ka philyoha-yess-ta. kulayse ku
 Mino-TOP some/one woman-NOM need-PST-DEC. so the
 yeca-lul manna-ss-ta.
 woman-PST meet-PST-DEC.
 ‘Mino needs a certain woman, so he met her.’

This shows that *etten* and *han* are specificity markers. In the following two sections, I will discuss their usages and find out when they are used as specificity markers.

2 *Etten* as a Specificity Marker

2.1 Syntactic status of *etten*

etten is believed to have come from a verb *etteh(a)*, which means ‘having some/what property’. This morphological change is generally observed in adnominal verbal endings, as shown below:

- | | | | | | |
|-----|-------|------------|--|-----------|-----------|
| (4) | khu | ‘be big’ | | khu-n | ‘big’ |
| | cak | ‘be small’ | | cak-un | ‘small’ |
| | talli | ‘run’ | | talli-nun | ‘running’ |

This allows the possibility that *etten* is just an adnominal modifier. However, there are expressions like *ku* ‘the’, *i* ‘this/these’ and *ce* ‘that/those’, which have

a fairly fixed position in relation to other adnominals in the NP structure, and *etten* has the same distribution as them.

- (5) a. *ku/i/ce/etten* *twu say cip*
 the/these/those/some two new houses
 ‘the/these/those two new houses’
- b. **twu/say ku/i/ce/etten* *say/twu cip*
 two(/new) the/these/those/some new(/two) houses
 ‘the/these/those/some two new houses’
- c. ?*ku/i/ce/etten* *say twu cip*
 the/these/those/some new two houses
 ‘the/these/those/some two new houses’

Here *twu* ‘two’ is a numeral and *say* ‘new’ an adjective of a basic form used as an adnominal. Notice that *etten* precedes a numeral or an adjective, just like *ku*, *i*, and *ce*, as pointed out in Suh (1996), Im (1987, 1998), etc. We can take them to be determiners which occupy the specifier position of a NP, leaving it unexplained how numerals and adjectives are ordered.^{2 3}

Furthermore, the other determiners cannot be used with *etten*, except *ku*.

- (6) **i/*ce/ku etten cip*

We posit a single position for a determiner, so this can be easily explained by assuming that *etten* is a determiner. *ku* can be used with *etten*, but there are some restrictions in its use. First, the combination must be used by being bound by a free choice operator.

- (7) ??*chelswu-nun ku etten say cha-lul sa-ss-ta.*
 chelswu-top the some new car-acc buy-pst-dec
 ‘Chelswu bought (the) some new car.’

²As pointed out in Kang (2001), one problem with taking them to be determiners is that they exceptionally take their complements to the right, unlike other categories in Korean, whose complements occur to the left. For this reason, we cannot say that they are heads of DPs.

³Adnominal clauses can occur in various positions. If they occur before definite determiners, they are interpreted as nonrestrictive modifiers, while if they follow determiners, they can be either restrictive or non-restrictive.

chelswu-nun ku etten cha-lul sa-to il nyen-ul mos
 chelswu-**top** the any car-**acc** buy-evev.if one year-**acc** not
 ssu-n-ta.
 use-**npst-dec**

‘Whatever new car he buys, Chelswu does not use it for one year.’

Here *ku etten* must be bound by the free choice binder *-to* ‘even if’. This indicates that *ku etten* is not definite, despite the expression *ku* ‘the’. It is more like an intensifier. Based on the discussion so far, we can claim that *etten*

2.2 Four uses of *etten*

There are four uses of *etten*. In Korean some indefinites are freely bound by an interrogative operator and used as a *wh*-word. In the following two examples, *etten* is used in interrogative sentences, and they are used as meaning ‘what kind/type of’, which ranges over properties.

- (8) a. A: mino-nun etten yeca-lul manna-ss-nya?
 mino-**top** what.property.of woman-**acc** meet-**pst-int**
 ‘What type of woman did Mino meet?’
 B: ippun yeca-lul manna-ss-ta.
 pretty woman-**acc** meet-**pst-dec**
 ‘(He) met a pretty girl.’
 b. A: ne-nun [_{NP} etten umsik]-ul cohaha-nya?
 you-**top** what.kind.of food-**acc** like-**int**
 ‘What (kind of) food do you like?’
 B: maywun umsik-ul cohaha-n-ta.
 spicy food-**acc** like-NPST-**dec**
 ‘(I) like spicy food.’

In these two examples, the questions with *etten* ‘what type/kind of’ are replied with some properties like *ippun* ‘pretty’ or *maywun* ‘hot’.

The use of a *wh*-word ranging over properties is the most common use of *etten*, but when it is forced, *etten* can be used as a *wh*-word ranging over individuals.

- (9) wuli pan-ey [_{NP} etten haksayng]-i ne-lul ccochatani-nya?
 our class-in which student-**nom** you-**acc** stalk-**int**
 ‘Which student in my class is stalking you?’

In this example, *wuli pan-ey* ‘in our class’ restricts the range of the individuals, and by the expression of *etten haksayng* ‘which student’ the speaker intends to ask which specific student is stalking the addressee.

In Korean, an indefinite can be used as a *wh*-word or an existential quantifier, depending on whether it is bound by an interrogative operator.

- (10) mino-nun [_{NP} etten yeca]-lul cohaha-n-ta-ko
 mino-**top** some.type.of woman-**acc** like-**npst-dec-cmp**
 malhay-ss-ta. (kulena, etten yeca-i-nci
 say-**pst-dec**. (but, what.type.of woman-be-whether
 kiekna-ci anh-nun-ta.)
 remember-**nml** not-**npst-dec**)
 ‘Mino said he likes some type of women. (But I don’t remember what type.)’
- (11) $\exists P[\text{said}(m, \wedge GENx[P(x) \wedge \text{woman}(x) \rightarrow \text{like}(c, x)])]$

etten in this example seems to range over properties, but it is supposed to refer to a particular property.⁴ However, *etten* here is not bound by an interrogative

⁴A similar phenomenon can be observed in an example of *nwukwu*. *nwukwu* is used as a *Wh*-word or as a non-specific indefinite in an interrogative clause.

- i. mino-nun nwukwu-lul cohaha-nya?
 mino-**top** who-**acc** like-**int**
 ‘Who does Mino like?’ or
 ‘Does Mino like anyone?’

However, it can be used as a specific indefinite.

- ii. a. nwukwu-nun coh-keyss-ta.
 someone-**top** good-**mod-dec**
 ‘Someone will be happy.’
 b. [x will be happy]^{g[x/d]}
- iii. a. mino-ka nwu(kwu)-ka o-ass-ta-ko malhay-ss-ta. (kulentey,
 mino-**nom** someone-**nom** come-**pst-dec-cmp** say-**pst-dec** (but
 nwukwu-i-nci kiek-i an na-n-ta.)
 who-be-whether remember-**nom** not come-**npst-dec**)
 ‘Mino said who came, but I don’t remember who it was.’

operator. So the NP gets existential meaning by existential closure, as given in (11). The property variable is bound by the existential quantifier, but the individual variable x is bound by the generic operator because the sentence is about people who have a particular property in general. This is a natural result from the fact that *etten* ranges over properties.

Finally, a NP with *etten* is used as an indefinite which has existential meaning and *etten* ranges over individuals.

- (12) a. mino-nun [_{NP} etten celmun yeca]-lul manna-ss-ta.
 mino-**top** some young woman-**acc** meet-**pst-dec**
 ‘Mino met some young woman.’
 b. $\exists x[\text{woman}(x) \wedge \text{young}(x) \wedge \text{met}(m,x)]$

In this example, *etten celmun yeca* is an indefinite which has existential meaning and ranges over female young individuals.

We have seen four usages of *etten*, and the next question is how *etten* is characterized as a specificity marker and how it is ensured to be used as a specificity marker. In general, we can say that a NP with *etten* can be used as specific, either denoting a specific property or referring to a specific individual. However, we generally assume that a specific indefinite refers to an individual. In addition, when *etten* ranges over properties, there is another operator which is bound by the generic operator. For this reason, we only consider a NP with *etten* a specific one when it ranges over individuals. So specific uses of NPs with *etten* can be ensured by excluding the possibility that they range over properties or they are bound by an interrogative operator. It is with an adnominal modifier that *etten* is not used as a *wh*-word, nor ranges over properties. This is illustrated below:

- (13) ne-nun [_{NP} etten yeyppun yeca]-lul po-ass-nya?
 you-**top** what pretty girl-**acc** see-**pst-int**
 b. $\text{said}(c, \{p \mid \exists x[\text{person}(x) \wedge p = \text{came}(x)]\})$

In (ii), by the use of *nwukwu*, the speaker refers to one of the hearers in the conversational context. In (iii), *nwukwu* is supposed to refer to a particular person. In this use, the speaker does not only say that Mino said that there was someone who came, but implicates that he knew who it was.

‘Did you see some pretty girl?’

In this example, the indefinite only ranges over individuals, and *etten* cannot be used as a *wh*-word even though the sentence is an interrogative: the sentence is construed only as a *Yes-No* question with the indefinite ranging over individuals. Comparison of the following examples also shows the fact that a adnominal modifier makes *etten* range over individuals. (15) can be followed only by (17).

- (14) mino-nun [_{NP} etten yenghwa]-lul cohaha-n-ta-ko
 mino-**top** some.kind.of movies-**acc** like-**npst-dec-cmp**
 malhay-ss-ta. (property/individual)
 say-**pst-dec**
 ‘Mino said he liked some (kind of) movie(s).’
- (15) mino-nun [_{NP} etten sulpwun yenghwa]-lul cohaha-n-ta-ko
 mino-**top** some sad movie-**acc** like-**npst-dec-cmp**
 malhay-ss-ta. (individual)
 say-**pst-dec**
 ‘Mino said he liked some sad movie.’

In (??), the indefinite has no adnominal modifier and is ambiguous: *etten* can range over properties or individuals. (14) can be followed by both (16) and (17). In (??), however, the indefinite has a adnominal modifier *sulpwun* ‘sad’ and *etten* only ranges over individuals.

- (16) ecey kulen yenghwa-lul po-ass-ta-ko ha-n-ta.
 yesterday such movie-**acc** see-**pst-dec-cmp** do-**npst-dec**
 ‘He said he watched such a movie yesterday.’
- (17) ecey ku yenghwa-lul po-ass-ta-ko ha-n-ta.
 yesterday the movie-**acc** see-**pst-dec-cmp** do-**npst-dec**
 ‘He said he watched the movie yesterday.’

In Korean, *kule(ha)n yenghwa* refers to a kind of movies, while *ku yenghwa* refers to a specific individual movie with definiteness. This shows that a NP with *etten* and a modifier ranges over individuals, and it refers to a specific individual movie.

3 *Han* as a Specificity Marker

3.1 Syntactic status of *han* as a specificity marker

In Korean there are three structures where a numeral word can be used. Their distributions have the following paradigm:

- (18) i. Num CN
ii. Num Cls-**pos** CN
iii. CN Num Cls

The first use can be characterized as a numeral occurring before a common noun, and the last two as a numeral occurring before a classifier or a unit word. These are illustrated as follows:

- (19) twu nonmwun
two paper
‘two papers’
- (20) twu pyen-uy nonmwun
two **cls**-of paper

‘two (units of) papers’
- (21) nonmwun twu pyen
paper two **cls**
‘two (units of) papers’

Common nouns and classifiers can be preceded by numerals and both can head NPs. For this reason, we take classifiers or unit words to be nouns, just like common nouns. However, I assume that classifiers or unit words have an additional feature like [+cls] and that nouns with the feature [+cls] are defective in that they do not assign a case to the preceding NPs.

- (22) ku nonmwun(*uy) sey pyen
Cf. ku sey peyn-uy nonmwun

In either case, there is an agreement which captures the co-occurrence between the noun and its appropriate classifier.

J.-S. Nam (2006) pointed out that numerals cannot freely occur before common nouns.

(23)	*sey kay	three dog		sey-mali-uy kay	three- cls -of dog
	*yel cangkwan	ten minister		yel-myeng-uy cangkwan	ten-cls-of minister
	*ney kkoch	four flower		ney-songi-uy kkoch	four-cls-of flower

However, small numbers like *twu* ‘two’, *sey* ‘three’ and *ney* ‘four’ are more likely to be allowed in front of common nouns which denote people.

(24)	twu/sey/ney/??tases/??yeses/... cangkwan
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This seems to be related to the possibility of being individuation. For this reason, they are not freely used in front of common nouns which denote things or animals.

- (25) a. ??twu/??sey/??ney/??tases kkoch ‘two/three/four/five flower’
b. ?twu/?sey/?ney/?tases kay ‘two/three/four/five dog’

If these are to be used appropriately, we need more than just existential interpretations of them: entities must be able to be individuated by attributing some properties to them. In this respect, *kkoch* ‘flower’ is less likely to be individuated than *kay* ‘dog’.

The specificity marker *han* can precede these nouns if other small numbers do so.

- (26) a. (?)han kay-ka mwun-ul twutuli-ess-ta.
one dog-**nom** door-**acc** knock-**pst-dec**
‘One dog knocked on the door.’
b. han cangkwan-i sathoy-lul ha(y)-ess-ta.
one minister-**nom** resignation-**acc** do-**pst-dec**
‘One minister resigned.’

- c. ?yenghi-nun han kkoch-ul kkek-ess-ta.
 yenghi-**top** one rose-**acc** pick-**pst-dec**
 ‘Yenghi picked one rose.’

In these examples, *han kay* and *han cangkwan* refer to a specific individual dog and minister respectively. The i

As we have seen in the discussion of *etten*, numerals can be all preceded by a determiner and have relatively fixed positions in relation to other modifiers of nouns. This is shown below.

- (27) salam-tul-i ku/etten han/twu/se yeca-lul ccochaka-ss-ta.
 person-**pl-nom** the/some one/two/three woman-**acc** follow-**pst-dec**
 ‘People followed the/some one/two/three woman/women.’

In this respect, they are all like numerals in general. In addition to the fact that only small numbers can occur before common nouns, however, there is another crucial formal difference between numerals before common nouns and numerals before classifiers or unit words. When small numbers occur before common nouns, we can put a pause after them. This can be compared with the fact that it is odd to put a pause between a numeral and a classifier or unit word.

I have shown that the uses of small numbers before common nouns are related to individuation. Then why do we consider *han* as a specificity, in particular? There is one property that makes *han* different from other numeral expressions: *han* always makes a NP indefinite. Consider the following examples.

- (28) twu namca-ka kongwen-ey iss-ta. twu namca-nun wuntong-ul
 two men-**nom** park-at be-**dec**. two men-**top** exercise-**acc**
 ha-ko iss-ta.
 do-and be-**dec**
 ‘There are two men at the park. The two men are doing exercise.’
- (29) ??han namca-ka kongwen-ey iss-ta. han namca-nun wuntong-ul
 one man-**nom** park-at be-**dec**. one man-**top** exercise-**acc**
 ha-ko iss-ta.
 do-and be-**dec**
 ‘There is one man at the park. The one man is doing exercise.’

Other numerals can be used to make NPs definite when they are used before common nouns, as in (28). But *han* is always indefinite, as in (29). The second NP *han namca* cannot refer to the one introduced in the first sentence. This shows that *han* is inherently indefinite. This is why I take *han* to be a specificity marker, distinguishing it from other small numbers, which can be definite or indefinite.

A NP with *han* followed by a CN must denote a specific individual, while a NP with *han* followed by a classifier must denote a nonspecific individual. In (30), the expression *amwu nonmwun-ina* ‘whatever paper (it is)’ expresses that the paper introduced by the indefinite is nonspecific. It goes well with the structure “CNP *han* cls”, but not with “*han* CNP”.

- (30) a. mino-nun amwu nonmwun-ina [_{NP} nonmwun han phyen]-i
 mino-**top** any paper-or paper one **cls-nom**
 philyohay-ss-ta.
 need-**pst-dec**.
 ‘Mino needs one paper, whatever it is.’
- b. ??mino-nun amwu nonmwun-ina [_{NP} han nonmwun]-i
 mino-**top** any paper-or one paper-**nom**
 philyohay-ss-ta.
 need-**pst-dec**.
 ‘Mino needs one_{spec} paper, whatever it is.’

One thing to add is that when a modifier comes together, *han* is always taken to be a specificity marker. In (31), *cha* can be taken to be a unit word or a common noun, so *han cha* ‘one car’ is likely to be understood as meaning either ‘a truckload (of something)’ or ‘a specific car’. In (32), the indefinite with *han* comes with an adnominal modifier *ppalkan* ‘red’ and the ‘truckload’ reading disappears.

- (31) wuli-nun [han cha]-ka philyoha-ta.
 we-**top** one car-**nom** need-**dec**
 ‘We need a truckload (of something).’
- (32) wuli-nun [han ppalkan cha]-ka philyoha-ta.
 we-**top** one red car-**nom** need-**dec**
 ‘We need some red car.’

This supports again that if *han* is used with a classifier or a unit word, the indefinite is nonspecific, while it is used with a common noun, the indefinite becomes specific. And an adnominal modifier ensures the specific interpretation of an indefinite with *han*. This reflects the distributional constraint that only common nouns allow adnominal modifiers. I have discussed two uses of *han* and when it is used as a specificity marker. The discussion can be summarized as follows:

- (33) a. (CN) *han* cls:
- i. *han* is simply a numeral.
 - ii. A pause cannot be put before *han*.
 - iii. An adnominal cannot follow *han*.
- b. *han* CN:
- i. *han* is used as a specificity marker.
 - ii. A pause can be placed before *han*.
 - iii. An adnominal can follow *han*.

4 Properties of the two specificity markers

In this section, I will discuss common properties and differences of the two specificity markers.

4.1 Similarities between the two specificity markers

In Korean, a quantifier in the position of the object does not come into the scope of the negation operator. On the other hand, nonspecific indefinites without any quantity expression can be negated.

- (34) mina-nun khikhun namca-lul manna-ci ahn-ass-ta.
 mina-**top** tall man-**acc-nml** not-**pst-dec**
 ‘Mina did not meet a tall man.’

The sentence means that there was no tall man Mina met. A indefinite with a specificity marker follows the pattern of quantifiers: it does not come into the scope of negation.

- (35) mina-nun etten/han khikhun namca-lul manna-ci ahn-ass-ta.
mina-**top** some/one tall man-**acc-nml** not-**pst-dec**
‘There was a tall man who Mina did not meet.’

The sentence does not mean that Mina met no tall man. The only way to negate a specific indefinite is to deny the (potential) use of a specific indefinite by using the contrastive topic marker. But this leads to a difference between *etten* and *han*. This will be discussed in the next subsection.

As Hintikka () pointed out, a specific indefinite can have narrow scope w.r.t. a quantifier. The same thing is observed with NPs with *etten* or *han*.

- (36) taypwupwunuy chinkwu-tul-i [_{NP} etten/han sosel]-ul thaykhay-se
most friend-**pl-nom** some/one novel-**acc** choose-and
ilk-ko iss-ta. (ambiguous)
read-and be-**dec**
‘Most friends (of mine) have chosen some/one novel and are reading it.’
- (37) yeca-tul-i motwu [_{NP} etten/han namca]-lul cohaha-n-ta.
woman-**pl-nom** all some/one man-**acc** like-**npst-dec**
(ambiguous)
‘The women all like some/one man.’

Here the indefinite with *etten* or *han* can have wide or narrow scope w.r.t. a quantifier. This is not what is expected of referential NPs.

Hintikka () claimed that even though a specific indefinite may have narrower scope than a quantifier, it has scope over epistemic operators. Yeom (1998) showed that this is not the case, and claimed that specific indefinites show epistemic relativity. This is what is observed in the uses of *etten* and *han*. In the following examples, an indefinite with *etten/han* may, or may not, stay within a belief context.

- (38) a. mino-nun [_{NP} etten/han khikhun yeca]-ka ccocha o-ko
 mino-**top** some/one tall woman-**nom** follow come-and
 iss-ta-ko mit-ess-ta.
 be-**dec-cmp** believe-**pst-dec**
 ‘Mino believed that a tall woman was following her.’
- b. kulena, twuy-ey-nun amwu-to eps-ess-ta.
 but back-at-**top** anybody-also not.exist-**pst-dec**
 ‘But no one was behind him.’
- c. kulena, kunye-nun cip-ey ka-ko iss-ess-ta.
 but, she-**top** home-to go-and be-**pst-dec**
 ‘But she was just going home.’

In (38a), a specific indefinite is used in a belief context, and in (38b), the existence of a tall woman who was following Mino is denied in the main context. This shows that the specific indefinite has narrower scope than the belief operator. In (38c), on the other hand, the pronoun *she* refers to the woman introduced in the belief context from the main context. So we can say that the specific indefinite has wide scope over the belief operator, but it is not necessarily the case. (38a) can be followed by the following sentence.

- (39) kulena, kukes-un namca-i-ess-ta.
 but it-**top** man-be-**pst-dec**
 ‘But it was a man.’

Here the pronoun *it* refers to the individual introduced in the belief context, but it does not presuppose that the woman introduced in the belief context exists in the actual world. So we have to say that the specific indefinite still stays within the belief context. This does not mean that the indefinite is nonspecific. If it were, it would be impossible for a pronoun in the main context to refer to the individual introduced in the belief context.

When an indefinite which occurs in a belief context is nonspecific, it cannot be referred to from the main context. This is shown in the following example.

- (40) ??mino-nun [nolan cha han tay]_i-ka philyoha-ta-ko
 mino-**top** yellow car one **cls-nom** need-**dec-cmp**
 sayngkakhay-ss-ta. kulayse kukes-ul pilye cwu-ess-ta.
 think-**pst-dec** so -**acc** lend give-**pst-dec**

‘Mino thought he needed a yellow car, so I lent it to him.’

Here *han* is used before a classifier and the indefinite is nonspecific. So *kukes* ‘it’ cannot refer to the yellow car from the main context.

We have seen that a specific indefinite can have narrower scope than quantifiers or the belief operator. We know that the belief operator is taken to quantify over possible worlds, so it is also a case of scope interaction. Another case of quantification over possible worlds is a conditional. If an indefinite with a specificity marker occurs in the antecedent clause of a conditional, it can be within or over the scope of the antecedent clause.

- (41) *ney-ka hoksi* [_{NP} *etten/han celmun yeca*]-*lul* *manna-myen*,
 you-**nom** possibly some/one young woman-**acc** meet-if
ilum-ul mwule po-ala.
 name-**acc** ask see-**imp**
 ‘If you happen to meet some/one young woman, ask her name.’

The sentence allows for the possibility that the addressee meets no young woman.⁵ This shows that specific indefinites with *etten* or *han* take part in scope interactions.

Indefinites with *etten* and *han* may have the scope of the main context or a belief context, but not that of a desire context.⁶ In Heim (1992), it is claimed

⁵But if some modifiers are added, then it is more plausible to interpret the NPs as specific.

- i. *ney-ka hoksi etten/han khi-ka cak-ko kwuyewun yeca-lul*
 you-**nom** possibly some/one height-**nom** small-and cute woman-**acc**
manna-myen, ilum-ul mwul-e.po-ala.
 meet-if, name-**acc** ask-try-**imp**
 ‘If you happen to meet {some/one} cute short woman, ask her name.’

This seems to be purely a contextual matter.

⁶If a NP with *etten* stays in the scope of a desire context, it ranges over properties.

- i. *na-nun kakkum etten yeca-wa kyelhonha-ko sip-ta-ko malha-ciman,*
 I-**top** sometimes some woman-with marry-**cmp** like-**dec-cmp** say-but
thukpyelhi sayngkakha-ko iss-nun yeca-nun eps-e.
 particularly think-and be-**adn** woman-**top** not.exist-**dec**
 ‘I sometimes say that I want to marry some type of woman, but I don’t have a particular woman in mind.’

that a desire context is defined based on a belief context. This means that if a presupposition is triggered in a desire context, it must be satisfied in the belief context. This is supported by the following examples.

- (42) mino-nun [_{NP} etten/han nalkun cha]-lul sa-ko sip-e hay-ss-ta.
 mino-**top** some/one old car-**acc** buy-and like-and do-**pst-dec**
 ‘Mino wanted to buy some/one old car.’
- (43) kulena {na/?ku}-nun kulen cha-ka eps-(ulci
 but {I/he}-**top** such car-**nom** not.exist-(**mod**
 molu-n)-ta-ko sayngkakhay-ss-ta.
 not.know-**npst**)-**dec-cmp** think-**pst-dec**
 ‘But I/he thought there {was no/might not be} such a car.’

In (42), a specific indefinite is introduced in Mino’s desire context. When the existence of an old car is denied in Mino’s belief context, as in (43), or when the possibility of its non-existence is asserted, the discourse becomes odd. However, the old car may not exist in the speaker’s belief. This is what is predicted by Heim’s claim. There is still a possibility that the relevant car exists in Mino’s desire context, but if it were possible, the discourse would be fine. The awkwardness of the discourse also indicates that specific indefinites cannot stay within desire contexts.

So far I have discussed some properties that the two specificity markers have in common. Here is the summary.

(44)

	etten	han	a(n)	some	a certain
negation	×	×	✓	×	×
conditional - narrow	✓	✓	✓	✓	×
quantifier - narrow	✓	✓	✓	✓	✓
belief context	✓	✓	✓	✓	✓
desire context	×	×	✓	✓	×

(For comparisons of *a(n)*, *some* and *a certain*, see Farkas (2002).)

4.2 Differences between *etten* and *han*

The two specificity markers have some properties in common because they have the meaning of specificity. However, their uses as specificity markers can be taken to be adaptations from their original meanings. Even if they changed into specificity markers, some of the aspects of their original uses are expected to remain in the meanings. We suppose that differences between *etten* and *han* come from their original uses.

First of all, *etten* is bound by an unselective binder like *tunci*, but *han* is not.

- (45) mina-nun [_{NP} etten/*han cakun inhyeng]-i-tunci mou-n-ta.
 mina-**top** some/one small doll-be-or collect-**npst-dec**
 ‘Mina collects whatever small doll (she finds).’
- (46) mino-nun kaki pan [_{NP} etten/*han haksayng]-ul manna-tunci
 mino-**top** self class some/one student-**acc** see-or
 insa-lul ha-n-ta.
 greeting-**acc** do-**npst-dec**
 ‘Whichever student in his class he may see, Mino says hello.’

The fact that a specific indefinite can be bound by an unselective binder is surprising, because free choice is the opposite notion of specificity. In some previous analyses like Yeom (1998), Farkas (2002), etc., a specific indefinite is a rigid designator.⁷ This analysis cannot be maintained in any sense because a NP with *etten* can be a bound variable. In general, a specific indefinite is not bound by an unselective binder, but it is not a necessary condition for being specific.

Another typical test for unselective binding is to see whether an expression is bound by the generic operator. However, neither of them is bound by the generic

⁷In Yeom (1998), rigidity is relativized to information states, so a rigid designator may be defined with respect to one information state, but not the others. This is necessary for proper names of imaginative figures. In addition, indefiniteness requires some indeterminacy, which is expressed by multiplicity of rigid references. See Yeom (1998) for detailed discussion of this. I tried to capture this multiplicity with multiple information states. However, the fact that a specific indefinite is bound by an unselective binder, multiplicity must be captured within an information state.

operator. Especially, when we know that a NP with *etten* can be unselectively bound, this is more surprising.

- (47) [_{NP} *etten/han celmun hakca*]-ka ttokttokha-ta.
 some/one young scholar-**nom** smart-**dec**
 ‘Some/One young scholar is smart.’
- (48) mino-nun pothong [_{NP} *etten/han cakun kukcang*]-ey ka-n-ta.
 mino-**top** usually some/one small cinema-to go-**npst-dec**
 ‘Mino usually goes to some/one particular small cinema.’

The first sentence is about a particular young scholar, not about young scholars in general, and the second sentence is about a particular small cinema. This indicates that specific indefinites are not bound by the generic operator. One possibility is that the generic operator is not an unselective binder, or that *-tunci* is an unselective binder which has a different mechanism from the one involved in the unselective binding by the generic operator.

A second difference is that *etten* implicates a lot of alternatives, while *han* does not. It has been pointed out that the use of a specific indefinite in a sentence does not imply that there is a unique individual that satisfies the proposition, as pointed out in Yeom (1998).

- (49) mino-nun ecey *etten/han yeca-lul* manna-ss-ta.
 mino-**top** yesterday some/one woman-**acc** meet-**pst-dec**
 ‘Mino met some/one woman.’

This does not exclude the possibility that Mino met another woman. However, it is also plausible that it was the only woman that Mino met yesterday. On the other hand, when a topic marker is attached to a NP with *etten* or *han*, it is implied that there is more than one individual involved.

- (50) [_{NP} *etten/han yeca*]-nun ccalpun paci-lul ip-ko iss-ess-ta.
 some/one woman-**top** short trousers-**acc** wear-and be-**pst-dec**
 ‘Some/One woman was wearing shorts.’

This may be related to the interpretation of the topic marker as contrastive: it yields a set of alternatives, which implies that there are other individuals than

the one which makes the sentence true. When the topic marker is used twice, the two specificity markers make different implications about the existence of other individuals.

- (51) [_{NP} etten/han yeca]-nun ccalpun paci-lul ip-ess-ko, [_{NP}
some/one woman-**top** short trousers-**acc** wear-**pst**-and
etten/han yeca]-nun kin chima-lul ip-ess-ta.
some/one woman-**top** long skirt-**acc** wear-**pst-dec**
‘Some/One woman was wearing shorts, and some/one woman a skirt.’

When we use NPs with *etten* twice, we assume that there are a lot of women in the context. On the other hand, with the two uses of NPs with *han* we can assume that there are only two women relevant in the context.

This may be related to another difference. A third difference is that *etten* seems to denote a certain property, even when the NP.

- (52) motun tali-ka etten/?han moyang-ul ha-ko iss-ta.
every bridge-**nom** some/one shape-**acc** do-and be-**dec**
‘Every bridge is in some/one shape.’

In general, every bridge is supposed to be in some shape. In this respect, the sentence should be uninformative. Even in this case, *etten* is natural. This may be because *etten* implies a certain property that selects a single shape differentiating itself from others. On the other hand, *han* does not require this because it does not require that there be other individuals. This example is compared with the following example.

- (53) motun tali-ka [etten/han thukihan moyang]-ul ha-ko iss-ta.
every bridge-**nom** some/one peculiar shape-**acc** do-and be-**dec**
‘Every bridge is in some/one peculiar shape.’

A modifier is added, and the sentence becomes fine even with *han*. *etten* contains a property even when it ranges over individuals.

As I have shown above, a specific NP is not normally negated by the negation operator. On the other hand, if a contrastive topic marker is used, the specific

indefinite is forced to go into the scope of the negation. In this case, the NP with *etten* can easily go into the scope of the negation operator and be negated. On the other hand, *han* robustly refused to be negated.⁸

⁸The cancellation of presupposition of existence when *etten* is used is similar to what is observed in presupposition cancellation in general.

- (1) ??mino-ka ku-uy kacok-ul manna-ci anh-ass-ta. ku-nun kacok-i eps-ta.
 mino-**nom** he-**pos** family-**acc** see-**nml** not-**pst-dec**. he-**top**
 family-**nom** not.exist-**dec**
 ‘Mino did not see his family. He has no family.’
- (2) mino-ka ku-uy kacok-ul manna-ci-nun anh-ass-ta. ku-nun kacok-i
 mino-**nom** he-**pos** family-**acc** see-**nml-top** not-**pst-dec**. he-**top**
 eps-ta.
 family-**nom** not.exist-**dec**
 ‘Mino did not see his family. He has no family.’

ku-uy kacok presupposes that Mino has a family, and the presupposition is not canceled without the contrastive topic marker *nun*. In the second example, the presupposition is cancelled with the help of the contrastive topic marker. In this case, the effect of the definiteness disappears: the denotation of the definite NP is more like a concept or notion which does not refer to anything in the actual world. The same thing is observed in the negation of a specific indefinite above: specificity is negated.

More generally, the use of a contrastive topic marker is related to scope relation. When a universal quantifier can have narrower scope than the negation, it is generally required that a contrastive topic marker occur between the negation and the universal quantifier.

- i. mino-ka motun chayk-ul ilk-ci ahn-ass-ta.
 mino-**nom** every book-**acc** read-**nml** not-**pst-dec**
 ‘Mino did not read every book.’
- ii. mino-ka motun chayk-ul ilk-ci-nun ahn-ass-ta.
 mino-**nom** every book-**acc** read-**nml-top** not-**pst-dec**
 ‘Mino did not read every book.’

In the first example, the sentence seems to have only one reading in which the universal quantifier has wide scope over the negation. To make the universal quantifier have narrower scope, we need the contrastive topic marker. This may make us think that the contrast topic marker forces a focused element to have narrower scope than the negation operator. However, there is one difference. The contrastive topic marker is a focus-sensitive. In (2), the definite description is focused, but the actual reading of the sentence is not about the alternative set of it. The negation simply negates the existence of Mino’s family. In (ii), on the other hand, the universal quantifier is focused and the sentence is about the alternative set of the quantifier. In (54), where the specific indefinite is focused, the sentence is not about the alternative set of the specific handsome man. The negation negates the existence of such a handsome man. In this respect, it is more like the presupposition failure.

- (54) mina-ka [etten/?han cal sayngkin namca]-lul sakwuy-ko
mina-**nom** some/one well shaped man-**acc** date-and
iss-ci-nun anh-ta.
be-**nml-top** not-**dec**
‘Mina does not date with some handsome man.’

This sentence should mean that there is no handsome man who Mina dates with. In this case, the indefinite with *etten* is fine, but the indefinite with *han* seems a little awkward. This is related to the fact that a NP with *etten* triggers a presupposition that can be cancelled, but the presupposition of existence triggered by *han* does not seem to be cancelled easily.

- (55) [etten/?han cengcikhān cangsakkwun]-i iss-nun kes-un ani-ta.
some/one honest dealer-**nom** be-**npst** thing-**top** not-**dec**
‘It is not the case that there is some/one honest dealer.’

In this example, the presupposition of the existence of an honest dealer is negated successfully when *etten* is used, but *han* does not seem to allow this. The use of *han* makes a robust presupposition of existence.⁹

The discussion so far gives the following implications:

- *etten* – not quantificational; *han* – quantificational $\Rightarrow$ Specificity does not

⁹Ionin (2006) claims that the use of a specificity marker *this* or *a certain* carries felicity conditions, instead of presuppositions. This does not seem to be the case. When an expression is used inappropriately, the use is denied by ... *ka anila* ... *ta* ‘not ... but ...’. A test using this expression does not seem to support that specificity markers carry felicity conditions.

- i. mino-ka cinci-lul tu-si-n kes-i ani-la pap-ul mek-ess-ta.
mino-**nom** rice_{hon} eat-**hon-rel**_{pst} thing-**nom** not-**dec** rice-**acc** eat-**pst-dec**
‘It is not that Mino ate_{hon} rice_{hon}, but that he ate rice.’
- ii. ?mino-ka etten/han celmun yeca-lul manna-n kes-i ani-la celmun
mino-**nom** some/one young woman-**acc** meet-**rel**_{pst} thing-**nom** not-**dec** young
yeca-lul manna-ss-ta.
woman-**acc** woman-**acc** meet-**pst-dec**
‘It is not that Mino met some/one young woman, but that he met a young woman.’

When Mino is not a senior, it is inappropriate to use an honorific expression in describing his action of eating meal. The first example expresses the inappropriateness. In the second example, the inappropriateness of specificity is expressed only to make the sentence odd. This indicates that specificity markers do not carry felicity conditions.

rely on whether or not the NP is quantificational.

- NPs with *etten* and *han* may have narrow scope with respect to other operators than the generic operator or the desire operator. $\Rightarrow$ Specificity does not depend on scopes.
- *etten* can be unselectively bound. $\Rightarrow$ *etten* introduces a variable unselectively bound, just like $a(n)$ in English.
- Given that *etten* introduces a variable unselectively bound, the fact that specific indefinites with *etten* are not bound by the generic operator or by the desire operator must be explained on the basis of types of values given to the variables.

It has been pointed out that specific indefinites have different properties from nonspecific indefinites. First, they tend to have wide scopes over other operators. It has been shown that they do so even when there are some syntactic barriers in between. However, they are not like proper names, as pointed by Hintikka (), Farkas (), Abusch (), etc. He showed that a specific indefinite may have narrower scope than a quantifier.

(56) Every boy in this room wants to date with a certain girl tomorrow.

Here, each boy has a particular girl in their mind, so the specific indefinite has narrower scope than the universal quantifier. This refutes the possible claim that indefinites are simply ambiguous between specific uses and non-specific uses. Abusch () even shows that indefinites can have intermediate scopes between two operators. Then how can indefinites have various scopes even without quantifier raising?

As an answer to this question, some scholars like Reinhart (1997), Winter (1997), Kratzer (1998), Matthewson (1999), etc. that indefinites quantify over choice functions. Geurts () and von Stechow () showed that they need a lot of modifications to capture the meanings indefinites show to the extent that we have to give up the idea. The problems can be avoided by assuming some sort of movement analysis, even if it is not a syntactic movement. Yeom (1997, 1998) propose that specific indefinites trigger presupposition of existence and that

presuppositions can have various scopes, as van der Sandt (1992), Yeom () and Geurts () claim.

However, the idea of scope taking is not enough to account for interpreting specific indefinites. There are some peculiar phenomena Ioup () showed indefinites have. I discuss two examples here.

- (57) Alberta believes that a dragon ate her petunias.
 = There is a particular dragon that Alberta believes ate her petunias.
- (58) Everyone_i believes that a_j witch blighted their mares.

In the first example, *a dragon* must have narrower scope than *believes*, but it seems to have a specific reading. This cannot be accounted for by scope relations. Ioup claims that the second example has four readings, which are summarized as follows:

- (59) everyone > a witch (i) specific (iii) nonspecific
 a witch > everyone (iii) specific (iv) nonspecific

Among the four readings, reading (iii) is not hard to catch within the standard semantics. Everyone thinks of a witch that she blighted their mares, but the speaker does not believe that the witch exists in the actual world. This means that *a witch* has scope over *everyone*, which in turn has wide scope over *believes*. But it has narrower scope than *believes*. The scope relations are circular as given below:

- (60) reading (iii): everyone > believes > a witch > everyone

This reading can be ascertained by the fact that (65) can be followed by (61).

- (61) If they ever find out who she is they'll try to catch her.

This shows that *a witch* is a specific indefinite. Even if so, the speaker does not believe that the witch exists in the actual world.

This reminds us of the famous Hob-Nob sentence, in which more than one person thinks about the same witch which does not actually exist. These examples cannot be explained simply by scope relations.

5 Specific indefinites ranging over individual concepts globally defined

To account for scopes of indefinites, a lot of analyses resort to syntactic operations or choice functions. However, the accounts are not satisfactory, and it may be that we can get the answer from a semantic approach. In this paper, I propose one semantic analysis. To account for the properties of the two specificity markers, I propose that a specific indefinite ranges over individual concepts. In standard dynamic semantics, a variable ranges over individuals, and the value of the variable is considered together with a possible world in which the value resides. On the other hand, individual concepts are defined with respect to (a subset of) possible worlds, and individuals involved in each individual concept are considered together as a whole. We have seen that a specific indefinite introduced in a belief context can be referred to by a pronoun in the main context. This indicates that wherever a specific indefinite is introduced, the value has global effect on the information state. To capture this effect, variables introduced by specific indefinites must be dealt with separately from ordinary variables which range over individuals. Moreover, Aloni (1997) proposes that ordinary variables for quantifiers also range over individual concepts. Following his analysis, I assume that all variables range over individual concepts, and specific indefinites are interpreted differently. To get the right results, I assume that a common ground consists of three components: a set of possible worlds, an assignment for ordinary NPs and an assignment for specific indefinites.

Let $M = \langle W, D, V, I \rangle$, where W is a set of possible worlds, D a set of individuals, V a set of variables and I an interpretation function for constants.

a set of individual concepts $IC = \{ic: U \rightarrow D \mid U \subseteq W\}$

$G = \{g: V' \rightarrow IC \mid V' \subseteq V\}$

common ground: a set of $\langle U, E, F \rangle$, where U is a subset of W , E a set of variable assignments for ordinary NPs ($E \subseteq G$), and F a set of variable assignments for specific indefinites ($F \subseteq G$).

We assume that variables range over individual concepts in which the individuals are (**epistemic counterparts**). Individual concepts are **partial** functions from possible worlds to individuals. (See Aloni (1997) and Gerbrandy (2000) for ‘individual concept’ or ‘epistemic counterpart’.) In a common ground, variable assignments for ordinary NPs and those for specific indefinites are dealt with separately.

Now I need to give interpretation rules together with rules for interpreting specific and nonspecific indefinites.

Given that $s = \langle U, E, F \rangle$,

$$s[P(x)] = (i) \langle U', E', F \rangle,$$

where $\forall w, g[w \in U \ \& \ g \in E \ \& \ g(x)(w) \in I(P)(w) \rightarrow w \in U' \ \& \ g \in E']$,

if $x \in \text{dom}(E)$

$$(ii) \langle U', E, F' \rangle,$$

where $\forall w, g[w \in U \ \& \ g \in F \ \& \ \forall w' \in U'[g(x)(w) \in I(P)(w)] \rightarrow w \in U' \ \& \ g \in F']$,

if $x \in \text{dom}(F)$

$$(iii) \text{ undefined otherwise}$$

$$s[\phi \ \& \ \psi] = s[\phi][\psi]$$

$$s[\exists_{ns} x \alpha] = \langle U', E', F \rangle \text{ such that for every } w \in U', \text{ there is an assignment } g \in E' \text{ such that } \langle \{w\}, \{$$

$$s[\exists_s x \alpha] = \langle U', E, F' \rangle \text{ such that for every } h \text{ in } F' \text{ and } w \in U' \cap \text{dom}(h), \langle \{w\}, E, \{h\} \rangle[\alpha] = \langle \{w\},$$

$$E+[x] = \{g \cup \{x \rightarrow ic\} \mid g \in E, ic \in IC, \exists w: ic(w) \in D\}, \text{ if } x \notin$$

$$\text{dom}(g) \text{ (undefined otherwise)}$$

$$F \oplus [x] = \{h \cup \{x \rightarrow ic\} \mid h \in F, ic \in IC \text{ for every possible world } w \text{ in } \},$$

$$\text{if } x \notin \text{dom}(h) \text{ (undefined otherwise)}$$

In standard analyses, a variable ranges over individuals, but in the analysis here, a variable always ranges over individual concepts. Instead, in a context $s = \langle U, E, F \rangle$, a variable x can be given an individual d by an individual concept ic_i in a possible world w_i , but it can be given another individual d' by a *different* individual concept ic_j in a different possible world w_j . On the other hand, a variable from a specific indefinite is given an individual by the *same* individual

concept across different possible worlds in a context.

This is illustrated in the following. Consider some individual concepts in which the individuals are epistemic counterparts. Suppose that the current context is compatible with w_1 , w_3 , w_5 and w_6 .

variable assignments for nonspecific indefinites

	w_1	w_2	w_3	w_4	w_5	w_6
ic_1	a	c	e	b	a	e
ic_2	b		a		c	a
ic_3	c	a			b	c

variable assignments for specific indefinites

	w_1	w_2	w_3	w_4	w_5	w_6
ic_1	a	c	e	b	a	e
ic_2	b		a		c	a
ic_3	c	a			b	c

ic_1 is defined with respect to w_1 through w_6 , while ic_2 is defined only with respect to w_1 , w_3 , w_5 and w_6 . And ic_3 is defined only with respect to w_1 , w_2 , w_5 and w_6 . For a variable x from a nonspecific indefinite, it can be given an individual c in a possible world w_1 by $g(x) = ic_3$, but it can be given an individual a in the possible world w_3 by $g(x) = ic_2$, and a and e in the possible worlds w_5 and w_6 , respectively, by $g(x) = ic_1$. Notice that a variable can be assigned an individual concept like ic_3 which is not fully defined in the current information state. On the other hand, for a variable y from a specific indefinite, it can be given b , a , c and a in the possible worlds w_1 , w_3 , w_5 and w_6 respectively by the same individual concept $g(x) = ic_2$.

5.1 Non-rigidity of reference

In some previous analyses, it is assumed that a specific indefinite is a rigid designator. (See Yeom (1997) and Farkas (2002).) However, this assumption leads to the wrong result. Consider the following example.

- (62) phathi-eyse bo-n etten/han yeca-ka mina-i-lci-to
 party-at see-**adn** some/one woman-**nom** mina-be-**mod-del**
 molu-ko ani-lci-to molu-n-ta.
 not.know-and not-**mod-del** not.know-**npst-dec**
 ‘Some/One woman I saw at the party might, and might not, be Mina.’

If a specific indefinite is a rigid designator, the proper name *Mina* should be a rigid designator. If two rigid designators are used in an identity statement, it becomes a necessity statement despite the possibility operator. Then the sentence should be contradictory. However, this is not the case. This shows that specific indefinites are not rigid designators. Individual concepts do not require the expressions in question to be rigid designators. Some individual concepts can be rigid individual concepts, but they are special.

5.2 From a property to an individual

We have seen that there are two variables involved in the meaning of *etten* when they are used as ranging over properties: one for a property and the other for an individual. However, when a NP with *etten* is used as a specific indefinite, the meaning can be captured only with a variable ranging over individual.

- (63) a. mino-nun [etten yeca]-lul manna-ss-ta.
 mino-**top** some woman-**acc** meet-**pst-dec**
 ‘Mino met some (type of) woman/women.’
 b. $\exists P \exists x [P(x) \wedge \text{woman}(x) \wedge \text{met}(m, x)]$ (nonspecific)
 c. $\exists x [\text{woman}(x) \wedge \text{met}(m, x)]$ (specific)

The question is where the variable for a property went in the specific use. A more plausible analysis is that the specificity marker maintains the variable for properties. This is possible when we assume that a specific indefinite with *etten* has a variable for a property which selects a unique individual in each possible world. One crucial difference from definite descriptions is that the individuals are taken to be counterparts across possible worlds.

If this analysis is on the right track, we can change the specific interpretation above into the following.

$$(63) \quad c'. \exists x[\exists P\forall y[P(y) \rightarrow y=x] \wedge \text{woman}(x) \wedge \text{met}(m,x)]$$

Here P must be the property that determines an individual concept which is defined with respect to a subset of possible worlds.

5.3 Ioup's (1977) two examples

Ioup (1977) mentioned a couple of examples which cannot be accounted for in standard interpretations of indefinites. Earlier, Marcus (1962) and Orenstein (1972) mentioned these problems.

- (64) Alberta believes that a dragon ate her petunias.
 = There is a particular dragon that Alberta believes ate her petunias.

In this example, no dragon exists in the actual world, but the indefinite has wide scope over the belief context. This is problematic when we try to account for the meaning on the basis of scopes.¹⁰

Another example is something like the following:

- (65) Everyone_i believes that a_j witch blighted their mares.

Ioup claims that (65) has four readings, which are summarized as follows:

- (66) everyone > a witch (i) specific (ii) nonspecific
 a witch > everyone (iii) specific (iv) nonspecific

¹⁰For these problems, Ioup proposes that an existential quantifier has two different readings.

... $\exists x$ has traditionally been given at least two different readings, one an existential reading and one a substitutional reading. On an existential reading of the particular quantifier, the variable in question would be said to range over a domain of ontologically existing objects referred to as the value of that variable. On the substitutional reading, we are concerned with the substituends of the variables which are linguistic objects, that is, the names of the things referred to. When the substitution involves noun phrases, the substituends will be individual constants. (Ioup 1977: 236)

However, his analysis does not seem to be right. Individual constants are syntactic entities, so the substitutional reading is really an interpretation.

To account for these examples, I need to give the interpretation rule for the belief operator.

$$\begin{aligned} \langle U, E, F \rangle [\text{Bel}(a, p)] &= \langle \text{Dox}(a, U), E, F \rangle [p] \\ \text{Dox}(a, U) &= \bigcup_{w' \in U} \{w \in W \mid w \text{ is compatible with what } a \text{ believes in } w'\} \end{aligned}$$

The variable assignments in the main context are inherited in the belief context. All variables are introduced in the main context, even if they are actually introduced in a belief context. It does not cause any problem because the individual concepts are defined with respect to (a subset of) the possible worlds of the belief context.

Suppose that a triple $\langle U, E, F \rangle$, where one assignment f in F gives the variable for the specific indefinite an individual concept ic_i as its value, as follows:

	w_1	w_2	w_3	w_4	w_5	w_6
ic_i			a	c	c	e
	MC			Alberta's BC		

MC: main context; BC: belief context

Since the individual concept is not defined with respect to the main context, the dragon does not exist in the actual world. On the other hand, it is defined with respect to every possible world compatible with what Alberta believes. Thus it has the effect of the specific indefinite having narrower scope than the verb *believes*. Nevertheless, the specific indefinite has the effect of having scope over the belief context so that it can be referred to by a pronoun in the main context. This is because even if w and g changes, f gives a fixed value for each of the pair: for each value for the subject, there is a fixed value for the specific indefinite. This allows the specific indefinite to be always accessible for anaphora. This shows that a pronoun refers to an individual concept, not an individual. In the main context, no dragon exists, but the individual concept for the dragon is accessible from the main context.

The other example can be accounted for in a similar way, but in this case there is one more condition to be imposed in order for the specific indefinite to be

accessible from the main context. Given a set of triples $\langle U, E, F \rangle$, we can derive the belief context for each individual quantified over by the universal quantifier.

$$\text{Dox}(g(i)(w), U) \text{ for each possible world } w \in U \text{ and for each individual } g(i)(w)$$

$$\langle U, E, F \rangle [\text{Bel}(\text{everyone}_i, \text{a witch blighted their mares})] = \langle U', E', F' \rangle \text{ such that for every } w \in U', \text{ for every } d \text{ in } \{g'(i)(w) \mid g \in E \ \& \ g' \in E+[i]\}, \langle \bigcup_{w \in U'} \text{Dox}(d, w), E, F \rangle [\text{a witch blighted their mares}] = \langle \bigcup_{w \in U'} \text{Dox}(d, w), E', F' \rangle$$

In the second example, $g \in E$ gives different values for the value of the universal quantifier and $h \in F$ may give a different individual concept for each pair of w and a value for the universal quantifier. In such cases, it has the effect of the specific indefinite having narrower scope than the universal quantifier. However, g and h are independent of each other, so it is possible that for different values of the universal quantifier, h gives the specific indefinite the same value: that is, h gives the same individual concept to the specific indefinite, regardless of different g 's, as in the table:

		w_1	w_2	w_3	w_4	w_5	w_6
$g(i) = d$	$h(j) =$				e	a	c
$g(i) = d'$	$h(j) =$				e	a	c
$g(i) = d''$	$h(j) =$				e	a	c
...	...	...					
		MC			g(i)'s BC		

Since the individual concept assigned to the specific indefinite is not defined with respect to the main context, it has the effect of the indefinite having narrower scope than the verb *believes*. On the other hand, for different values assigned to the variable i for the universal quantifier, the variable j for the specific indefinite has the fixed value, which has the effect of giving the specific indefinite wide scope over the universal quantifier.

5.4 Anaphora over belief contexts

We have seen an example in which a specific indefinite is introduced in a belief context and a pronoun in the main context refers back to it.

- (67) mina-nun [etten/han khi-ka khun yeca]-ka ccocha o-ko
mina-**top** height-**nom** some/one tall woman-**nom** follow come-and
iss-ta-ko mit-ess-ciman, sasil namca-i-ess-ta.
be-**dec-cmp** believe-**pst-but**, fact man-be-**pst-dec**
‘Mina believed that some tall woman was following him, but in fact, it
was a man.’

How is this possible? This is possible because even if the specific indefinite is introduced in a belief context, the value assigned to it is defined with respect to the main context, as follows:

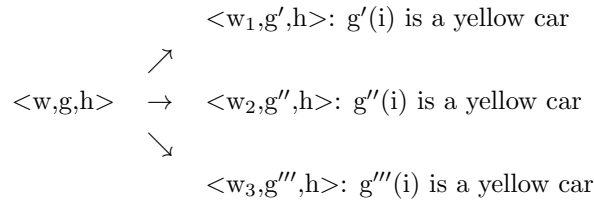
	w_1	w_2	w_3	w_4	w_5	w_6
ic	a	b	a	c	c	e
	MC			Mino's BC		

Even if this individual concept is introduced in a belief context, it is defined with respect to the main context, so it is fine to refer to it from the main context. For this reason it behaves as if it is introduced in the main context. It is possible that the specific indefinite introduces an individual concept which is defined with respect to all the possible worlds in the belief context, but not to all of the possible worlds in the main context. Then such individual concepts are removed from the context.

On the other hand, if a non-specific indefinite is introduced in a belief context, it is not referred to from the main context, as shown below:

- (68) ??mino-nun [nolan cha han tay]_i-ka philyoha-ta-ko
mino-**top** yellow car one **cls-nom** need-**dec-cmp**
sayngkakhay-ss-ta. kulayse kukes-ul pilye cwu-ess-ta.
think-**pst-dec** so -**acc** lend give-**pst-dec**
‘Mino thought he needed a yellow car, so I lent it to him.’

This can be easily explained if we assume that a non-specific indefinite introduces just an individual. If an individual is introduced in a belief context, It does not reside in any possible world in the main context. Moreover, for a possible world in the main context, there are many doxastic alternatives in which an individual is introduced. More specifically, for a triple $\langle w, g, h \rangle$ in the main context, w_1 , w_2 and w_3 are Mino's doxastic alternatives and g' , g'' and g''' are extensions of g .



Here each yellow car does not exist in the actual world. Moreover, $g'(i)$, $g''(i)$ and $g'''(i)$ are not counterparts. For these reasons, it is not possible for a pronoun in the main context to refer to the yellow cars.

5.5 Specific indefinites in desire contexts

We also have seen that a specific indefinite cannot stay within a desire context. This is observed in the following. Here (42) can be followed by (43), only when the subject is first person singular.

- (42) mino-nun [_{NP} etten/han nalkun cha]-lul sa-ko sip-e hay-ss-ta.
 mino-**top** some/one old car-**acc** buy-and like-and do-**pst-dec**
 ‘Mino wanted to buy some/one old car.’
- (43') kulena {na/?ku}-nun kulen cha-ka eps-(ulci
 but {I/he}-**top** such car-**nom** not.exist-(**mod**
 molu-n)-ta-ko sayngkakhay-ss-ta.
 not.know-**npst**)-**dec-cmp** think-**pst-dec**
 ‘But I/he thought there {was no/might not be} such a car.’

This is rather easily accounted for by the fact that an individual concept is a function from a set of possible worlds to a set of individuals which are epistemic

counterparts. In the example above, an *ic* must be defined with respect to Mino's belief context. The existence of the book can be denied only in the speaker's belief context, not in Mino's belief context. When Mino does not have a specific car in mind and he wants to have a specific car, the sentence becomes odd, as shown below:

- (69) ??mino-nun acik sayngkahay twu-n cha-nun eps-ciman,
 mino-**top** yet think put-**adn** car-**top** not.exist-but,
 etten/one nalkun cha-lul sa-ko sip-ta.
 some/one old car-ACC buy-**cmp** like-**dec**
 'Mino does not have a car in mind yet, but he wants to buy some/one
 (particular) old car.'

This shows that specificity itself does not become part of desire.

5.6 Specific indefinites in generic sentences

Next thing to explain is why a specific indefinite is not bound by the generic operator. Consider the following example.

- (70) $\phi =$ A lawyer is amoral.
 'For all x: if x is a typical/normal lawyer, x is amoral.'

This sentence means that generally normal lawyers are amoral. This illustrates a case where an indefinite is bound by the generic operator.

A specific indefinite does not lead to the same interpretation, as follows:

- (71) A certain lawyer is amoral.

This can be explained if a variable introduced by a specific indefinite ranges over individual concept. Suppose that x is an individual concept ic_i , where individuals involved are epistemic counterparts.

	w_1	w_2	w_3	w_4	w_5	w_6
ic_i	a	b	a	c	c	e
	normal			not normal		

If a variable ranges over individual concepts, the individuals in each individual concept must be dealt with simultaneously. If individual concepts are to be bound by the generic operator, each and every individual of them must be normal lawyers. However, the individuals are epistemic counterparts. They are not selected as normal lawyers. Thus the individuals cannot all be normal lawyers. All in all, the generic operator only binds variables which range over individuals, not those ranging over individual concepts. And individual concepts do not constitute a domain for the generic operator.

5.7 Specific indefinites with narrow scope

We have seen that a specific indefinite can have narrower scope than a quantifier.

- (72) yeca-tul-un motwu_i [etten/han_j namca]-lul kitali-n-ta.
 woman-**pl-top** all some/one man-**acc** wait-**npst-dec**
 Every_i woman is waiting for some/one_j man.

Suppose that the sentence is evaluated with respect to $\langle U, E, F \rangle$, where $w_3 \in U$, $g \in E$ and $h \in F$.

		w_1	w_2	w_3	w_4	w_5	w_6
$g(i)(w_3) = d$	$h(j) =$	a	b	a	c	c	e
$g'(i)(w_3) = d'$	$h(j) =$	b	c	e	a	b	a
$g''(i)(w_3) = d''$	$h(j) =$	c	a	a	e	a	c
...	...	...					
				MC			

In the table, for each different assignment, we get different individuals for the universal quantifier over women, and different men in the world w_3 . This has the effect of the universal quantifier having wide scope over the specific indefinite. Even though the specific indefinite is assigned different individual concepts, we only consider the individuals in w_3 . It is no harm to have individual concepts in capturing the narrow scope of a specific indefinite.

Here the relative scope of i and j is captured by dependency relation between $g(i)$ and $h(j)$. In my analysis, the two assignments E and F are independent of

each other. It does not mean that they have to be independent. The possibility of being dependent is still open. Furthermore, the notion of an individual concept makes it easy to capture a functional relationship. Consider the following example:

- (73) Every_i true Englishman adores a_j certain woman—his mother. (Hintikka 1986)

This sentence means that in a world w , ‘for every $g(i)(w)$ which is a true Englishman in w , $g(i)(w)$ adores $h(j) = g(i)$ ’s mother and $h(j)$ is a woman in c ,’ (where $g(i)$ ’s mother is an individual concept). This indicates that even when a specific indefinite has narrow scope, the notion of individual concept is necessary to capture functional readings.

6 Conclusions

In Korean, a NP with a specificity marker can be bound by an unselective binder, but it is not bound by any binder. This leads us to assume that a specific indefinite has a different semantic type than ordinary noun phrases. From this, I propose that specific indefinites range over individual concepts. An individual concept is one which exists across possible worlds in a context. This means that the agent in such an information state knows a set of properties which can be attributed to the individual (s)he (thinks (s)he) knows. The agent is pretty sure that the attributions can choose a unique individual. This is compatible with using the notion of individual concept in capturing a specific indefinite. This does not mean that the agent can discern the individual among others. It is possible that the agent is wrong in making such an assumption: there may be more than one individual that has the properties (s)he attributes. Then it is a change of a domain of individuals. This is quite a different matter.

The uniqueness of an individual concept allows us to refer to it by a pronoun in a later discourse, even when the specific indefinite is introduced in a belief context. Moreover, the uniqueness can be relativized to a quantifier, so it is quite natural that a specific indefinite can have narrower scope than another quantifier. This is quite different from a case where a nonspecific indefinite has

narrower scope. Individual concepts give us more values than those that are ascertained in the actual world. This allows us to capture functional readings.

It is intuitively plausible that nonspecific indefinites range over individuals. I can give interpretation rules based on this assumption, but one problem with this analysis is that it is hard to keep embedded contexts like belief contexts in the representation of an information state. Geurts () proposed an idea to keep belief contexts for anaphora within belief contexts from later discourse, but the idea is complex and is not intuitively plausible. In my analysis, I separate world information and discourse information. The payoff is that nonspecific indefinites range over individual concepts, which is more complex than an analysis in which they range over individuals. But it gives us the idea that we can attributes various properteis to the individual, which means that we do not know which properties it has. This is what we want to capture in dynamic semantics.